Chavan Dinesh Karthik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	5.36
T.S.R JUNIOR COLLEGE	Class XII (TSBIE)	2023	91.00%
ST JOSEPH HIGH SCHOOL	Class X (SSC)	2021	10.00%

SKILLS

Programming Languages : C, Python, Matlab, HTML, CSS

OS : Windows, Linux

Softwares : LTspice, Latex, Electric, Git, Github, SolidEdge

POSITIONS OF RESPONSIBILITY

Diesta Hospitality Coordinator : Assisted with event management and coordinated hospitality arrangements.

RELEVANT COURSES

Core Courses : Circuit and Network Analysis, Signals and systems, Digital Systems, Electromagnetics, Control Systems, Semiconductor Device Fundamentals, Analog Electronics, Analog Circuits, Power Systems, Power electronics, electrical machines, digital signal processing, Information Theory, Intro to CMOS processing, VLSI, Integrated Circuits for Wireless Communications, Communication Systems, Wireless communications.

Lab Courses : Electrical Circuits Lab, Electronic Devices Lab, DSP Lab.

Math Courses : Calculus, Matrix theory, Probability and random processes, Differential Equations, Vector Calculus, Complex Variables.

PROJECTS

5-Bit Successive Approximation (SAR) ADC :

- Built a complete 5-bit SAR ADC using discrete components and verified its operation using an oscilloscope.
- Designed the system using an R-2R ladder,DAC,LM358 comparator, and 7474 flipflops for Analog to Digital conversion.
- Created a 1Hz clock using an OP-Amp stable multivibrator for stable and synchronized operation.

Mobile Charger Circuit Design :

- Designed a 5V,20W mobile charger by converting 230V AC to low-voltage DC.
- Used a step down transformer, bridge rectifier, filter capacitor, and voltage regulator.
- Demonstrated practical understanding of rectification, filtering and voltage regulation.

Discrete Hardware Digital Clock :

- Designed a 12 hr digital clock using D flipflops and asynchronous counters.
- Implemented mod-10 and mod-60 counters along with NAND -gate reset logic.
- Displayed time using 7 segment displays without using an microcontroller.

Adaptive Noise Cancellation System :

- Developed a real time noise cancellation system in MATLAB using LMS, NLMS, and RLS algorithms.
- Improved signal quality by reducing noise and achieving around 40dB SNR improvement.
- Verified performance using FFT and time domain analysis under different noise conditions,

Todo List Web App :

- Created a simple To-Do List application where users can add, complete, and delete tasks.
- Added filters to view all tasks, pending tasks, and completed tasks.
- Designed a responsive and user friendly interface using HTML, CSS and JavaScript.

Calculator :

- Developed a basic calculator to perform addition, subtraction, multiplication and division.
- Added features such as decimal number input and clear fucntionality.
- Implemented keyboard support for easier usage.
- Built the user interface using HTML and CSS and handled calculations using javascript.

EXTRACURRICULAR

- Participated in District-Level Kabaddi, Kho-Kho and Cricket competitions.
- Active NSO cricket player.